

HAMMAD

Karachi, Pakistan • mhammadfazil14@gmail.com • github.com/Hammad7-dot • hammad7-dot.github.io/hammad-portfolio

SUMMARY

AI & Data Science graduate (SMIT) who ships full working products, not notebooks — FastAPI backends, deployed ML pipelines, and scheduled ETL systems. Four solo-built, end-to-end AI applications covering multimodal ML, data engineering, and API design. Comfortable owning a project from spec to deployment, and documents technical trade-offs and bug fixes honestly rather than glossing over them.

SKILLS

Languages & Core: Python, SQL, Git

ML / Data: Pandas, scikit-learn, ANN (artificial neural networks), multimodal pipeline design (tabular + text + image)

Backend & Data Engineering: FastAPI, SQLAlchemy, PostgreSQL, Redis, ETL pipeline design, APScheduler, REST API design

Tools & Deployment: Streamlit, Docker & Docker Compose, Streamlit Community Cloud, Postman

PROJECTS

CrediShield — AI Co-Pilot for Bank Loan Underwriting | *Python, Streamlit, multimodal ML pipeline*

- Built a multimodal underwriting tool that reads an applicant's tabular financial data, free-text letter of intent, and KYC documents in one panel, and returns an automated risk score, charts, and a generated PDF summary.
- Cut an unreliable image-verification feature rather than ship it broken, then refactored the remaining pipeline into a clean multi-file architecture; deployed live on Streamlit Community Cloud.

Healthcare AI Co-Pilot | *Python, Streamlit, multimodal ML (imaging, tabular, NLP)*

- Built a 3-modality clinical decision-support panel: chest X-ray pneumonia screening, an ANN-based cardiac risk model on structured patient data, and a clinical notes review tab, unified in one review flow.
- Designed the imaging tab to accept real uploads or fall back to synthetic demo generation, with UI disclaimers making clear which parts of the demo are illustrative versus production-ready.

Dynamic ETL Weather Dashboard 2.0 | *Python, Streamlit, SQLAlchemy, OpenWeather API, APScheduler*

- Built a 3-stage ETL pipeline (extract → transform → load) that pulls live weather for 6 cities from the OpenWeather API hourly via a background scheduler, storing results in SQLite and surfacing them across a 5-tab live dashboard.
- Diagnosed and fixed 3 production bugs: a validation-ordering crash affecting missing-data handling, a TypeError from unhandled null timestamps, and duplicate scheduler instances caused by per-session (not per-process) state in Streamlit.
- Added automatic retry/backoff on transient API failures and threshold-based webhook alerts that fire once per event rather than every hourly cycle.

Real-Time Crypto Analytics Platform | *Python, PostgreSQL, Redis, FastAPI, Streamlit, Docker*

- Built a 7-stage ETL pipeline from a written spec: retry/backoff CoinGecko extraction, validation with dead-letter logging for rejected records, batched Postgres upserts, and a 5-minute scheduler.
- Extended the core pipeline with Redis-cached reads, a FastAPI REST layer, historical trend queries, Z-score anomaly detection, and a one-command Docker Compose stack.
- Found and fixed a silently incorrect KPI (a metric computing COUNT instead of AVG, mislabeled in the UI), logging the root cause and fix as a tracked engineering decision rather than a quiet patch.

EDUCATION

AI & Data Science Diploma — SMIT (Saylani Mass IT Training), Karachi